



Harvard John A. Paulson
School of Engineering
and Applied Sciences

CS 2880: AI for Social Impact, Fall 2024

RN-SOAR:

**Robust Nurse Scheduling in the Online Setting
using Adaptive Reinforcement Learning**

Jessica Chen, Alina Yu, Elizabeth Zhong

Advised by Prof. Milind Tambe and Shreshth Verma



Harvard John A. Paulson
School of Engineering
and Applied Sciences

WHERE
SCIENCE
AND
ENGINEERING
CONVERGE

Problem Statement

Nurse Scheduling and Exhaustion

Nurse exhaustion is a well-studied problem in the healthcare industry. Only 27.2% of nurses got at least six hours of sleep before shifts; 62% report experiencing burnout.

In this project, we address nurse exhaustion as it specifically links to scheduling problems.

- **Fatigue.** *Min et al (2018)*: Schedule characteristics such as overtime, shifts, and rest periods have strong impact on nurses, causing “overwhelming sense of tiredness and exhaustion.”
- **Understaffing.** *Lasater et al (2020)*: Medical-surgical units staffed on average 3.3-9.7 patients per nurse (vs. recommended 4); 49.1% report unfavorable hospital safety grade.
- **Flexibility.** *Dhaini et al (2018)*: 32% of nurses report limited or no say in planning their own shift schedules; work schedule flexibility were negatively related to emotional exhaustion.



Problem of Non-Robust Scheduling

To guide our work, we conducted interviews with nurses and nursing supervisors, who have worked in both hospital settings and clinic practices.

Existing systems fail to account for last-minute absences and fluctuating demand of care.

- Kaiser Permanente, one of the largest clinic systems in the world, integrates Epic healthcare systems with internal staffing tools. Despite Epic's staffing features, the schedule is finalized and set 3 hours before shifts start and patient demand is finalized.
- In the face of absences, the software does not go back and reroute people.
- Patient care demand is based on naive last-period projections and patient volume rates.

In the face of last-minute changes in need, supervisors must manually re-staff.

- Managers must text each person individually, "it's lucky if 1-2 people respond."
- Options for last minute staffing: Request for standby nurses, swap shifts, offer overtime, or remain understaffed.



Schedulers in Literature

Canonical literature addressing the Nurse Scheduling Problem are not robust.

- Exact optimization methods (integer programming) lack flexibility in handling real-life uncertainties
- Heuristic approaches (genetic algorithms, simulated annealing) offer more adaptability but struggle with large disruptions and are mostly limited to local solutions.
- Some studies have proposed stochastic optimization to introduce randomness in predicting events like absences, but is less relevant in the online, reactive setting.

RL methods have more recently been introduced as a robust online scheduling solution.

- *Zhang et al (2024)*: Showed that an MDP and Q-network metaheuristic approach outperforms traditional local-search-based metaheuristic algorithms.
- *Dinh et al (2024)*: Formulated the Online Dynamic Home Health Care Scheduling Problem (where nurses accept/reject patient requests for home visits) using an MDP RL approach.



Contribution: RN-SOAR

1. Formulate the nurse scheduling problem as a Markov Decision Process.
2. Devise a simple optimal algorithm to automate the process for re-staffing shifts in the face of last-minute absences or demand spikes.
3. Design and train a reinforcement learning model to provide robust and extendable scheduling services.





Harvard John A. Paulson
School of Engineering
and Applied Sciences

WHERE
SCIENCE
AND
ENGINEERING
CONVERGE

RN-SOAR: Methodology

Using Real-World Instances

We used a dataset provided by *Ceschia et al (2023)* that includes **34 real-world instances** from various healthcare institutions in North Italy.

Dataset Characteristics.

- Environment: Includes shift details, full-time and part-time contract terms (max shifts, min rest period, etc), nurse details (shift compatibility), time-off requests, and more. 30-day period.
- Simulated Annealing Schedule: Staffs nurses on shifts for the 30-day period.

→ To isolate our impact on addressing robustness to schedule last-minute changes (and simulate modular improvement to legacy schedulers), we formulate our MDPs based on a simulated annealing solution.



Benchmarking against Optimal Heuristic

To verify the performance and optimality of our RL model, we benchmark it against a simple optimal solution. We limit the scope of this project to single-swap solutions.

Optimal 1-Swap Heuristic:

If there is an absence, the scheduler can deterministically try several options:

1. Staff a nurse on standby.
2. Staff a nurse on a resting day that has not fulfilled contract shifts.
3. Offer overtime to nurses who are not currently staffed.
4. Allow the shift to remain unstaffed.



Defining the MDP

State	Transition Probabilities	Actions	Reward Function
Matrix of size $n \times m$, where n is the number of days and m is number of nurses. 0: Absent 1: Working 2: Standby 3: Resting - Remaining shifts - Overtime	Each “episode” is a 30-day period, or a single run of a data instance. At each day, there is a probability p_A of a nurse absence. In the case of an absence, another nurse may accept swap or overtime requests with probability p_O.	Given an absence, the model may: 1. <u>Swap shifts</u> between the absent nurses and a standby or resting without overtime nurse. 2. <u>Offer overtime</u> to a resting nurse. 3. <u>Remain understaffed.</u>	Rewards are assigned accordingly: <i>Swap success: +10</i> <i>Swap fail: -1</i> <i>Overtime success: 3</i> <i>Overtime fail: -1</i> <i>Unchanged: -5</i> In results, we show rewards are controlled for hyperparametric bias.



Implementation

- We train our RL model using the **Proximal Policy Optimization** algorithm.
Rationale: After an update, PPO's new policy should be not too far from the old policy. This matches our setting, where new schedules should not be too far off from existing.
- The RL model is implemented on *stable-baselines3*.
- Training is done against the same instance (a certain set of nurses working at a hospital).
- Convergence occurs over episodes (runs of 30-day periods).

Repository: <https://github.com/JessSanChen/RN-SOAR>

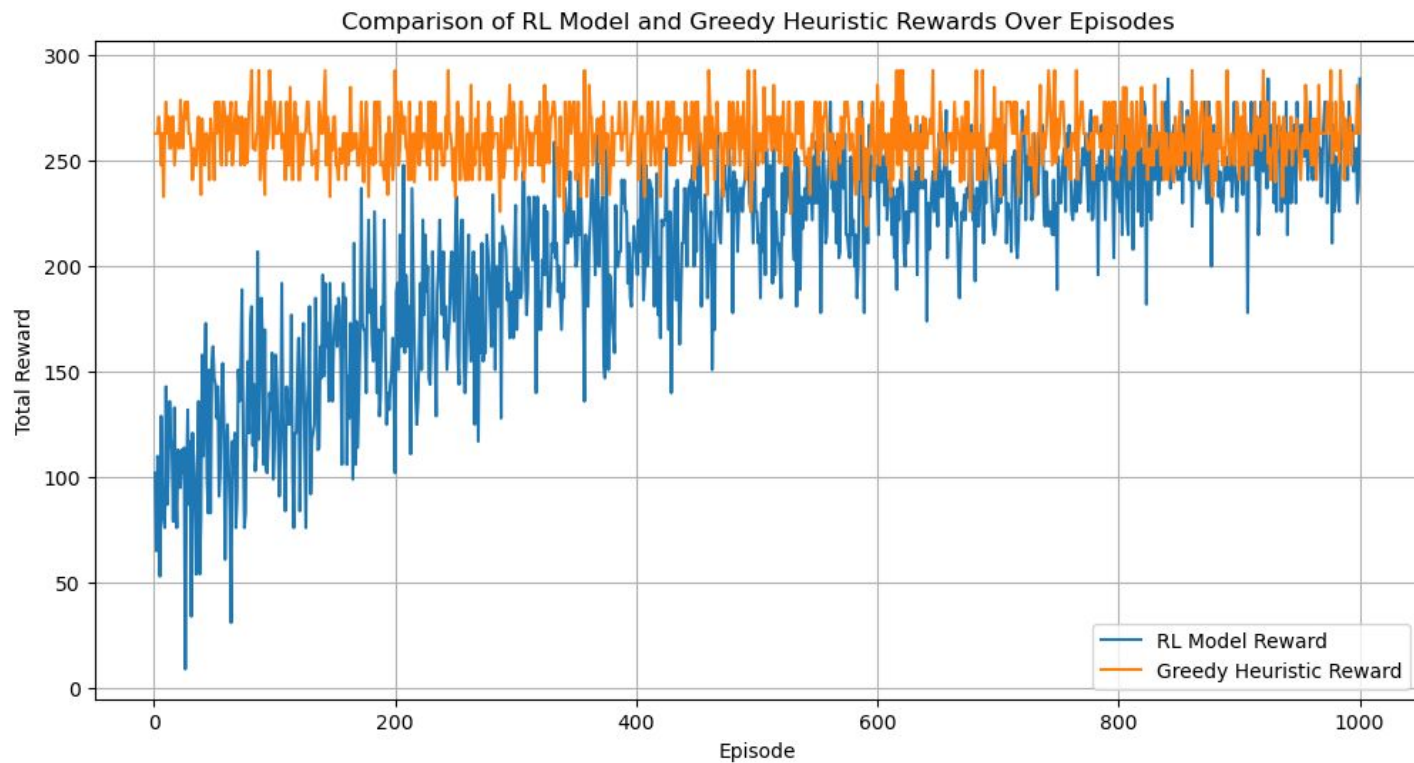




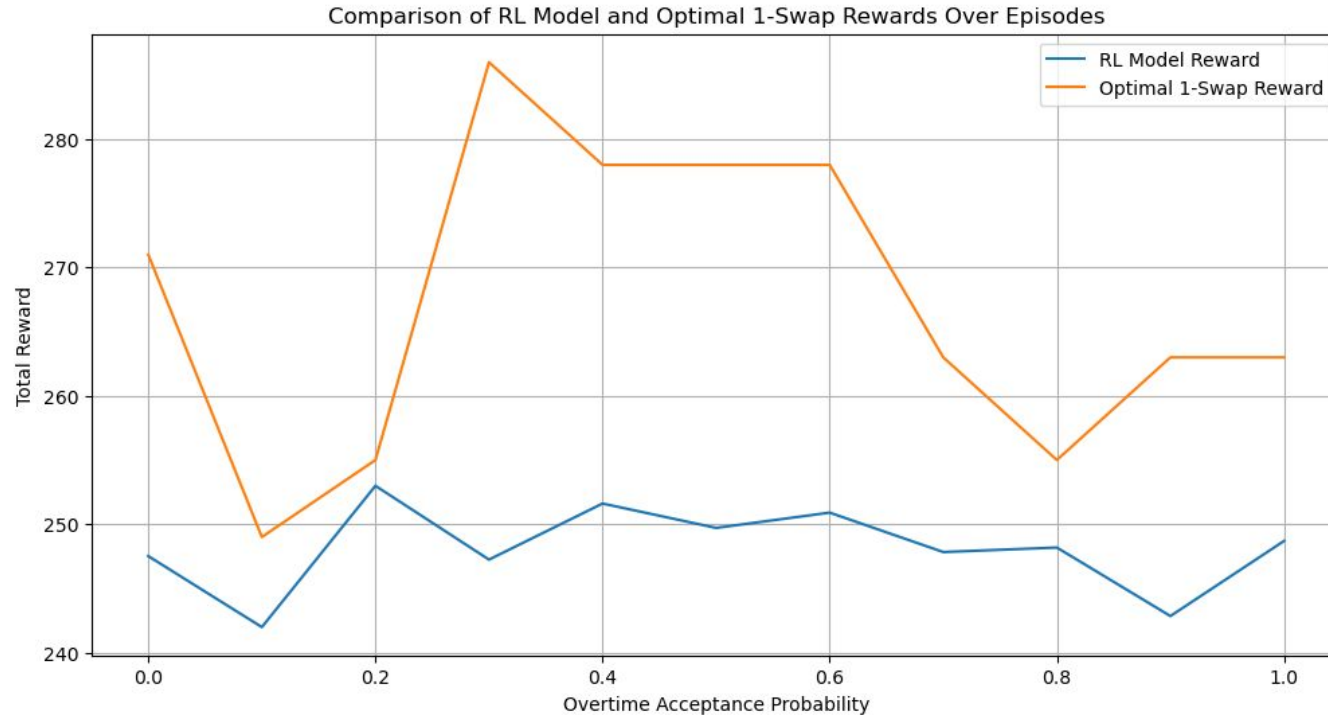
Harvard John A. Paulson
School of Engineering
and Applied Sciences

WHERE
SCIENCE
AND
ENGINEERING
CONVERGE

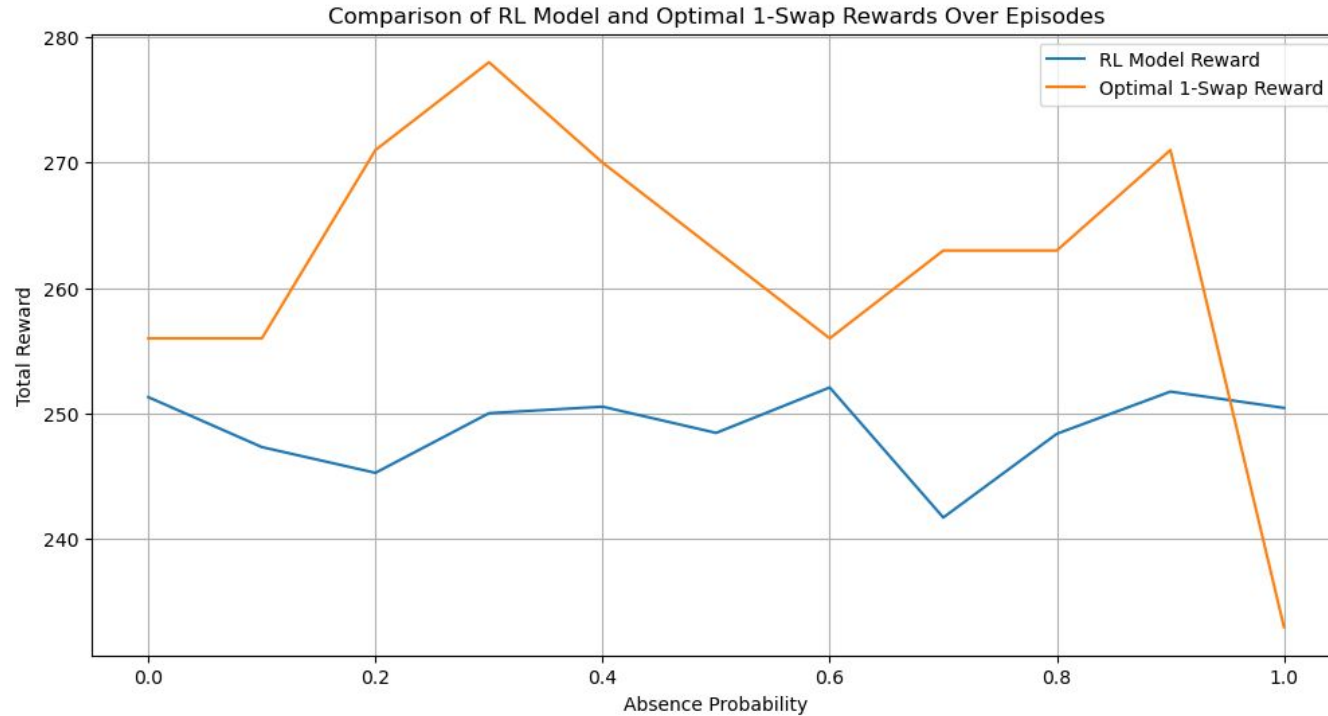
Results



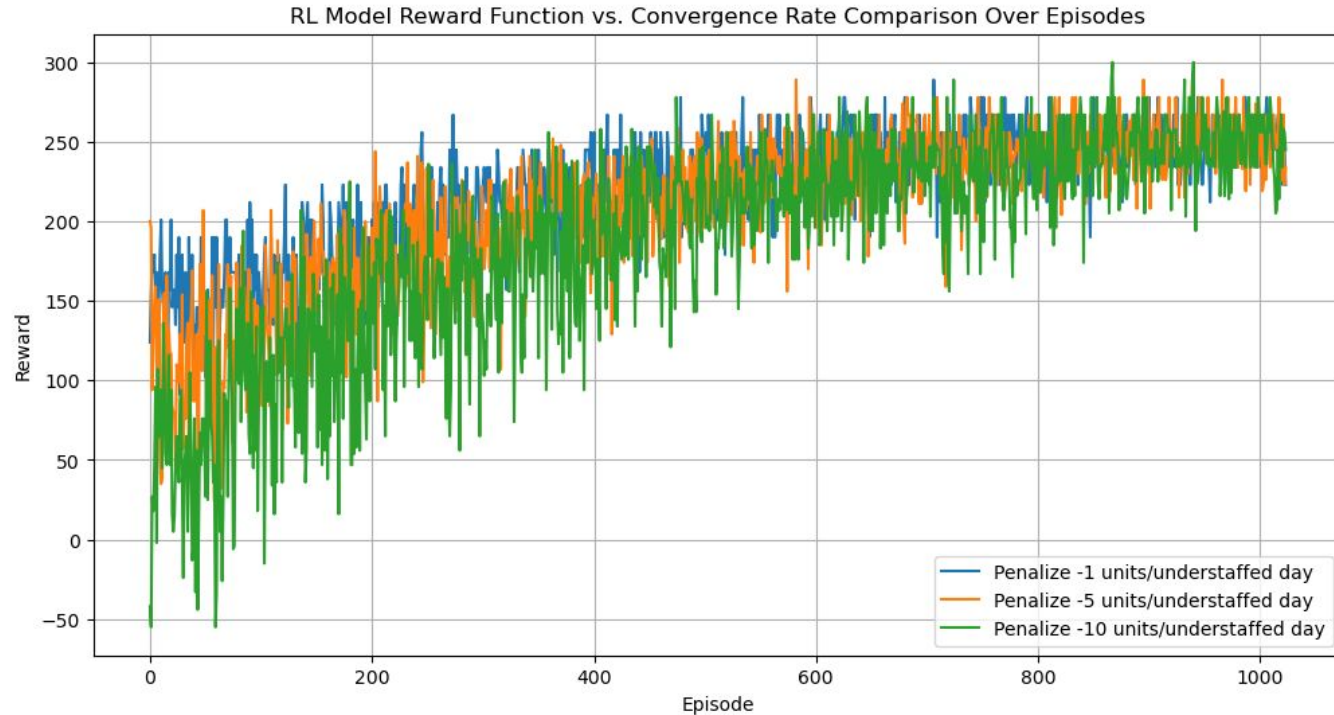
Post-convergence, RN-SOAR performs consistently, controlled for overtime acceptance probability hyperparameter.



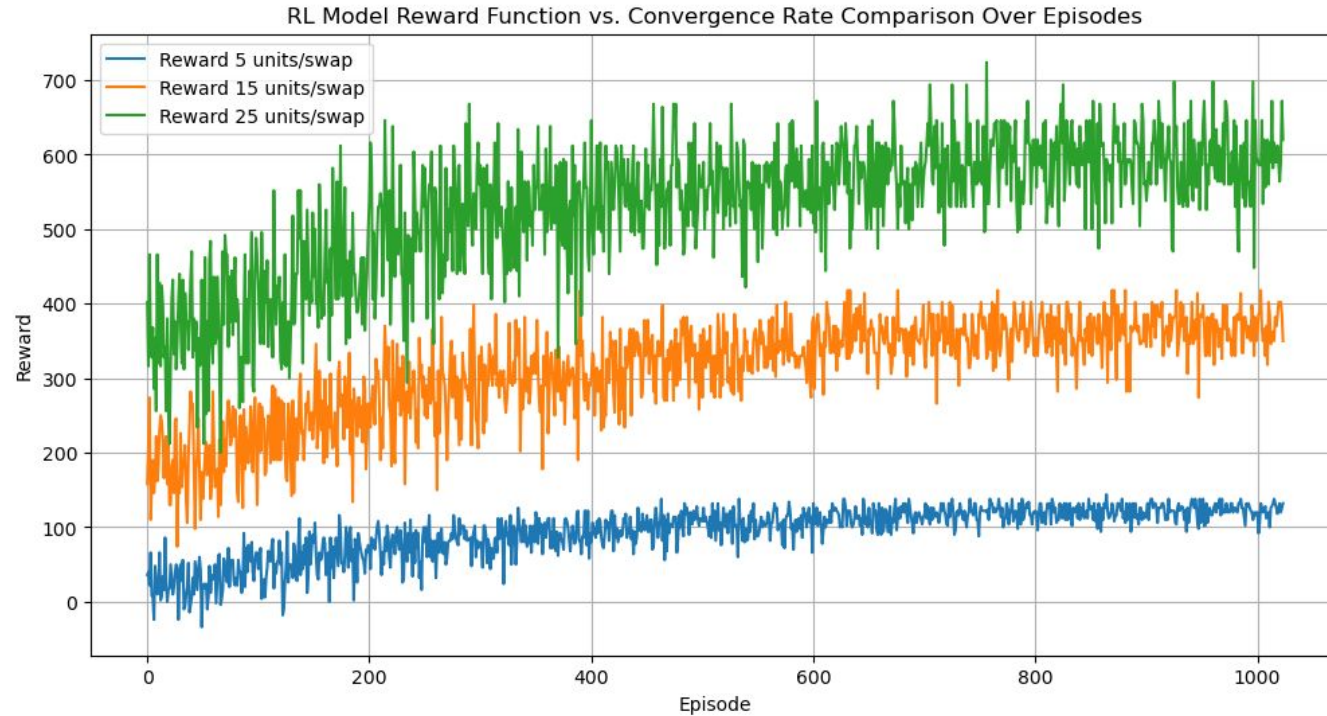
Post-convergence, RN-SOAR performs consistently, controlled for nurse absence probability hyperparameter.



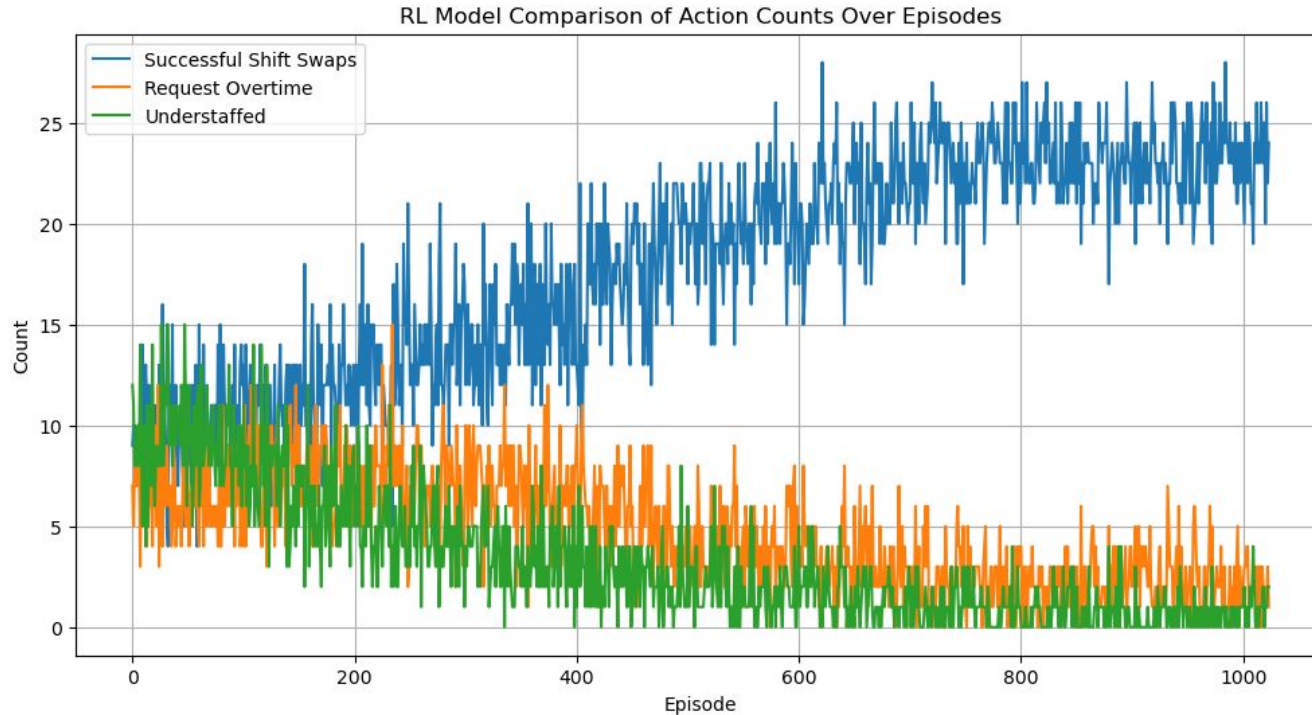
RN-SOAR converges at relatively the same pace, controlled for understaffing reward penalization hyperparameters.



RN-SOAR converges at relatively the same pace, controlled for successful shift swap reward hyperparameters.



RN-SOAR convergence to optimal is driven by the policy learning to favor successful utilization of standby/available nurses over relying on overtime or allowing understaffing.





Harvard John A. Paulson
School of Engineering
and Applied Sciences

WHERE
SCIENCE
AND
ENGINEERING
CONVERGE

Lessons Learned

Lessons Learned

- It's critical to consider the real life circumstances and the unexpected events that come up in nurse scheduling to create an effective practical solution
- Reinforcement learning provides a flexible, robust response to unexpected events.

Next Steps:

- Incorporate more constraints, such as skills-to-department matching.
- Augment action space to find multi-swaps.
- Optimize for efficiency, especially with exponentially larger action spaces.





Harvard John A. Paulson
School of Engineering
and Applied Sciences

WHERE
SCIENCE
AND
ENGINEERING
CONVERGE

Questions?